16. CONSTRUCTION OF SQUARES AND TRIANGLES

This chapter deals with the brainteasing problems of construction of squares by combination of three parts after selecting them from the list of five different alternatives numbered from A to E. The following discussion would assist us in solving such problems

Select a piece which contains a right angle between two adjacent outer edges Try to fit another piece in its hollow spaces. If you can't, select another piece. Repeat the procedure with different sets of such pieces. Finally with the two pieces fitting into each other, find the third piece which fits into the other two selected ones, to get a completed square finally. „

We now discuss a couple of solved examples

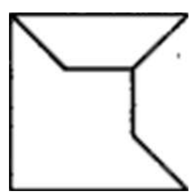
Example 1 :

Select three out of the following five alternative figures which together form one of the four alternatives (a), (b), (c) or (d) and when fitted together will form a complete square.

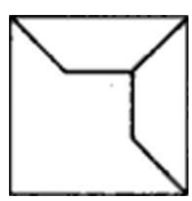
A B C D E

(a) ACD *(b)* CDE (c) BCD (*d)* ACE

Solution : The only figure with a right angle is fig. (C). Fig. (B) fits Into it as shown :

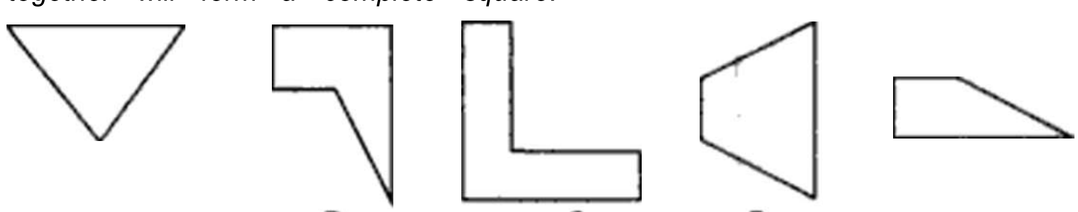


Finally, fig. (D) completes the square by fitting into the above combination
The completed square appears, as shown



a Figures (B), (C) & (D) will together form a square.
Hence, alternative (C) is the answer.

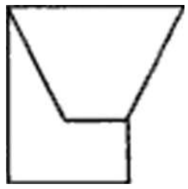
Example 2 : *Select three out of the following five alternative figures which together form one of the four alternatives (a), (b), (c) or (d) and when fitted together will form a complete square.*



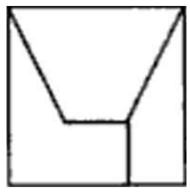
A B C D

(a) ACD **(*b*) BDE** **(*c*) ABD** **(*d*) ADE**

Solution : We begin with choosing a figure having a right angle. Rg **(A)** does not have any right angle. Rg **(B)** has a right angle. Now. we try to fit other pieces in fig. **(B).** We get fig (D) fitting into it; as shown :



We finally select fig. (E) fitting into this combination to complete the square; as shown ;

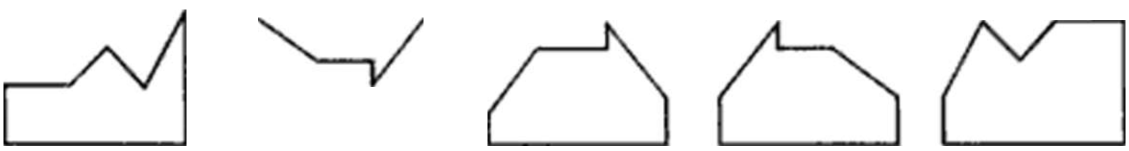


Figures **(B),** (D) & (E) together form a square.

Hence, alternative *(b)* is the answer.

A yet another type of problems on construction of squares is discussed below, in the following example.

Example 3 *Given below Is a set of five alternative figures marked (A), (B), (C)₉ (O) end (E). Select the figure which does not fit Into any of the remaining alternative figures to form a complete square.*



A **B** C **D** E

Solution : Clearly, fig. (A) fits into fig. (E) to focm a complete square and also. fig. **(B)** fits into fig. **(D)** to form a complete square as shown :

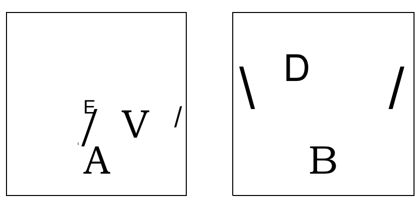
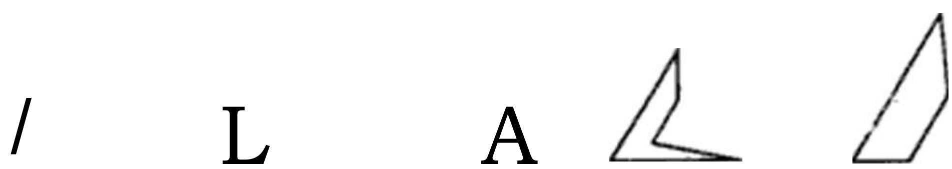


Fig. (C) does not fit in any of the alternative figures to form a square. Therefore, fig. (C) is the answer.

Similar to the construction of squares, we have problems on construction of equilateral triangles. The solving of such proWems will become easier after studying the following example.

Example 4 :



A B C D E

(a) ABD *(b)* ACE (c) ACD *(d)* CDE

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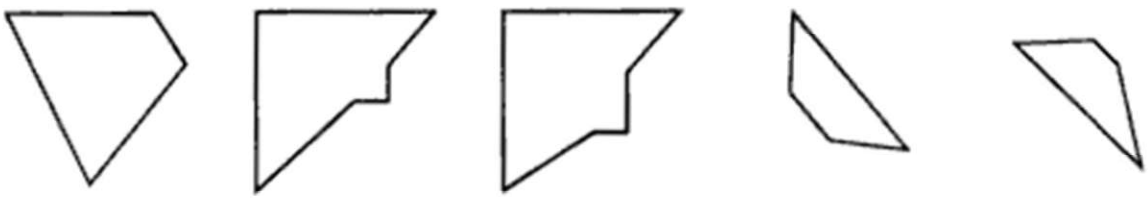
(A) (B) (C) (D) (E)

(a) ABC (b) BCD (0 CDE (d) BDE

(Hotel Management. 1993)

(A) (B) (C) (D)

(a) ACD (b) ADE (0 ABE (d) ABD



(A) (B) (C) (D) (E)

(a) ABC (b) BCD (0 ACD (d> CDE

(Hotel Management, 1993)

(A) (B) (C) (D) (E)

(a) ABE (b) ABC (c) BCE (tf) BCD

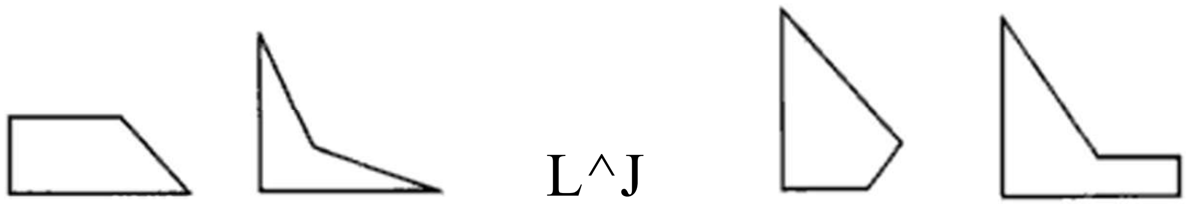
(A) (B) (C) (D) ^ (E)

(a) ABD <b) BCD (0 CDE (CO BCE

(Hotel Management, 1993)

(A) (B) (C) (D) (E)

(a) ABO (b) ABE (c) BCD (d) BDE



(A) (B) (C) (D) (E)

(a) ACD (b) ABD (0 BCD (d) CDE

(M.B.A. 1994)

(A) (B) (C) (D) (E)

(a) ADE (b) ACE (*c*) BCE (d) BCD

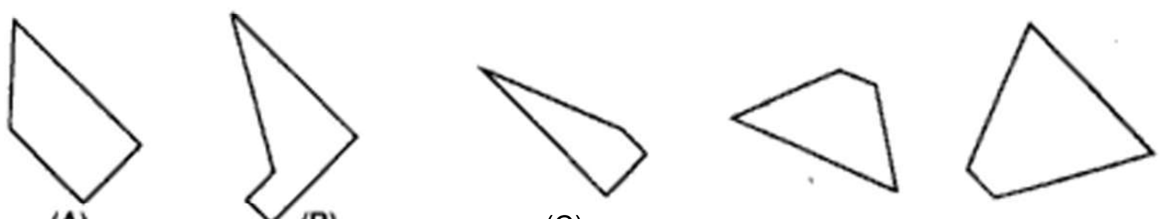
(A) (B) (C) (D) (E)

(a) ABC (b) BCD (*d*) ACD (d) CDE

(Hotel Management, 1993)

W) (B) (C) (D)

(a) ABC (b) ACD (c) BCE (d) CDE



(A) (B) (C) (D)

(a) ABD (b) BCE (c) ACD (di BDE

(Hotel Management. 1993)

» **Directions :** *In questions 29 to 33, five alternative figures, marked (A), (B), (C), (D) and (E) are given. From these five figures, we can get two pairs of figures which form squares. You have to select the odd figure which does not fit In any of the other alternative figures to form a complete square.*

(D) (E)

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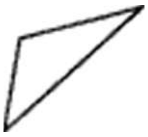
^ t i l ^  

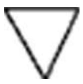
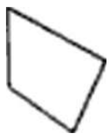
(A) (B) (C) (D) <E)

(a) ACE (b) ABD (c) BDE (d) CDE

45.    Λ

(A) (B) (C) (D) (E)

(a) ABC (b) ACE (C) BCD (d) BDE

46.  

(A) (B) (C) <D) (E)

(a) BCD (b) ABD (c) CDE (d) ABE

47. - A <3 O 

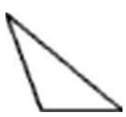
(A) (B) (C) (D> (E) ..

(a) ABD (b) BCD (0 BDE (d) CDE

48.  4 A 

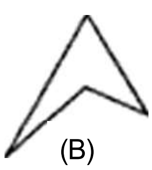
(A) (B) (C) (D) (E)

(a) ABC <b) ACE (C) BDE (efi CDE

49.   J \ O

(A) (B) <C) (D) (E)

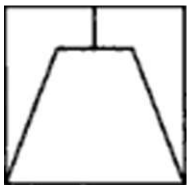
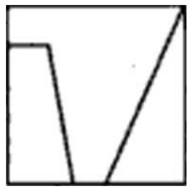
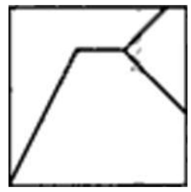
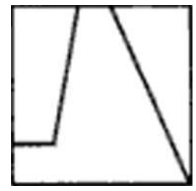
(a) BCD (b) ACD (0 CDE (d) BDE

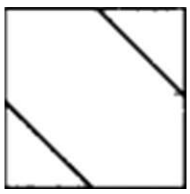
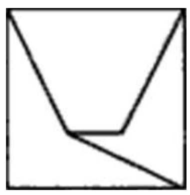
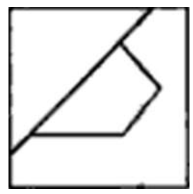
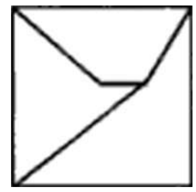
50.   

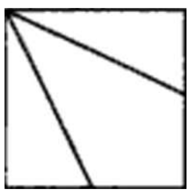
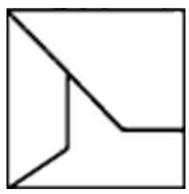
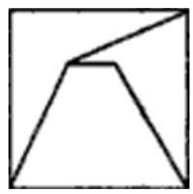
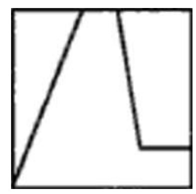
(A) (B) (C) (D) {E>

(a) ABC (b) BCD (c) ABD (d) ABE

ANSWERS

1. (d) 
2. (b) 
3. {b) 
4. (d) 

5. (d) 
6. (c) 
7. (d) 
8. (0 

9- (C) 
10. (c) 
11. (d) 
12. (c) 

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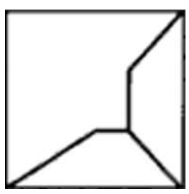
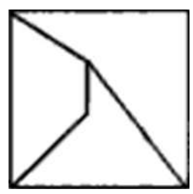
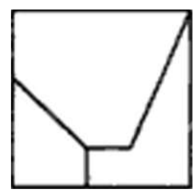
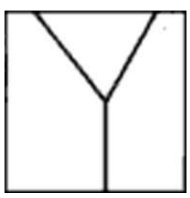
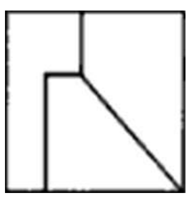
14. (d) 
15. (a) 
16. (b) 

17. (b) 
18. (c) 
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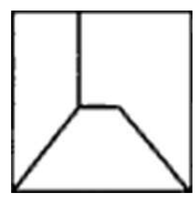
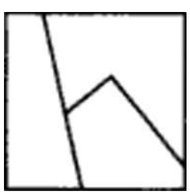
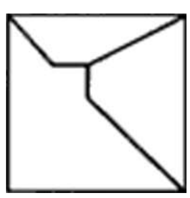
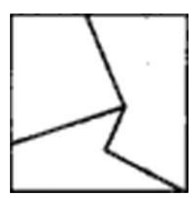
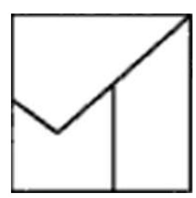
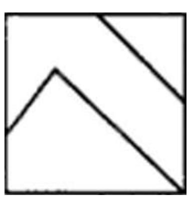
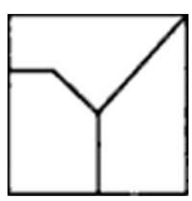
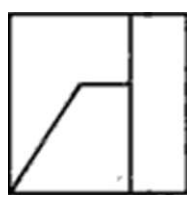
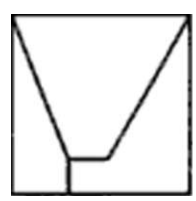
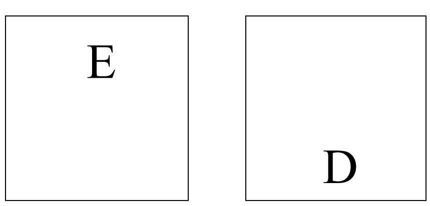
20. (d) 

21. (a) 
22. (a) 
23. <>) 
24. (a) 

25. <d) 
26. (a) 
27. (b) 
28. (0) 

29. (c) 

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